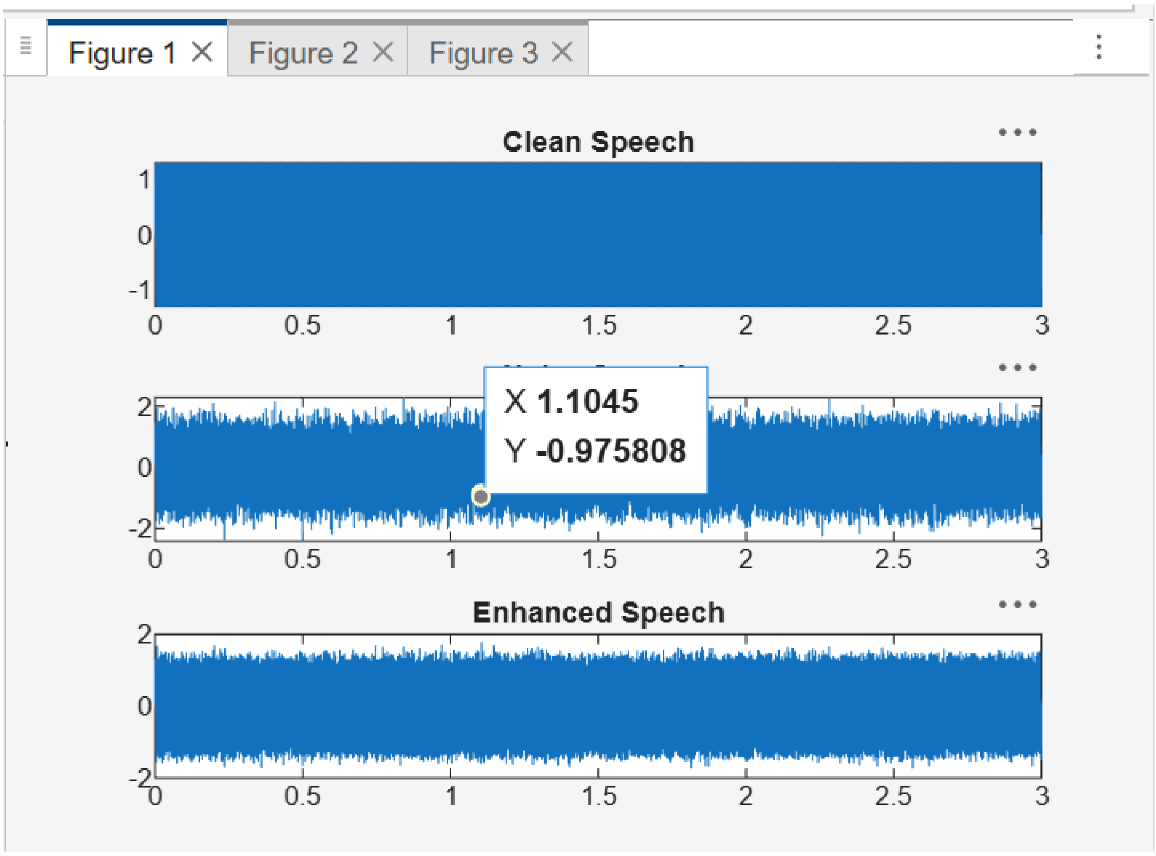
10.1 Implementation of Spectral Subtraction for Speech Noise Reduction

```
clc; clear; close all;

Fs=8000; t=0:1/Fs:3;
x=sin(2*pi*200*t)+0.5*sin(2*pi*400*t); % Clean signal
xn=x+0.3*randn(size(x)); % Noisy signal

Y=fft(xn);
N=fft(0.3*randn(size(x)));
xe=real(ifft(max(abs(Y)-abs(N),0).*exp(1j*angle(Y))));

% Signals
ffgure;
subplot(311),plot(t,x),title('Clean Speech')
subplot(312),plot(t,xn),title('Noisy Speech')
subplot(313),plot(t,xe),title('Enhanced Speech')

% Spectrograms
ffgure;
subplot(311),spectrogram(x),title('Clean Spectrogram')
subplot(312),spectrogram(xn),title('Noisy Spectrogram')
subplot(313),spectrogram(xe),title('Enhanced Spectrogram')

% Magnitude Spectra
```

```
figure;
subplot(311),plot(abs(fft(x))),title('Clean Spectrum')
subplot(312),plot(abs(fft(xn))),title('Noisy Spectrum')
subplot(313),plot(abs(fft(xe))),title('Enhanced Spectrum')
```

RESULT



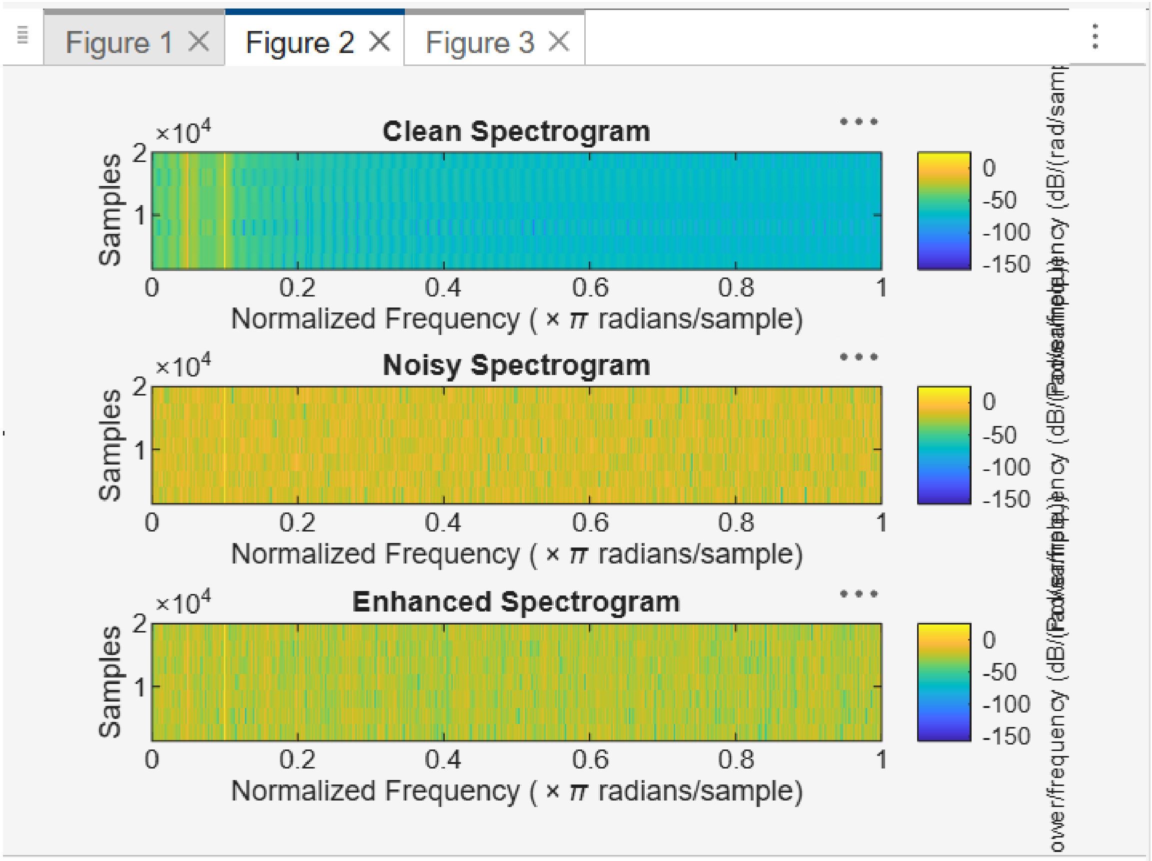


Figure 1
Figure 2
Figure 3
Clean Spectrogram
Noisy Spectrogram
Enhanced Spectrogram
Samples
Normalized Frequency (× π radians/sample)
×10^4
0
0.2
0.4
0.6
0.8
1
2
-50
-100
-150

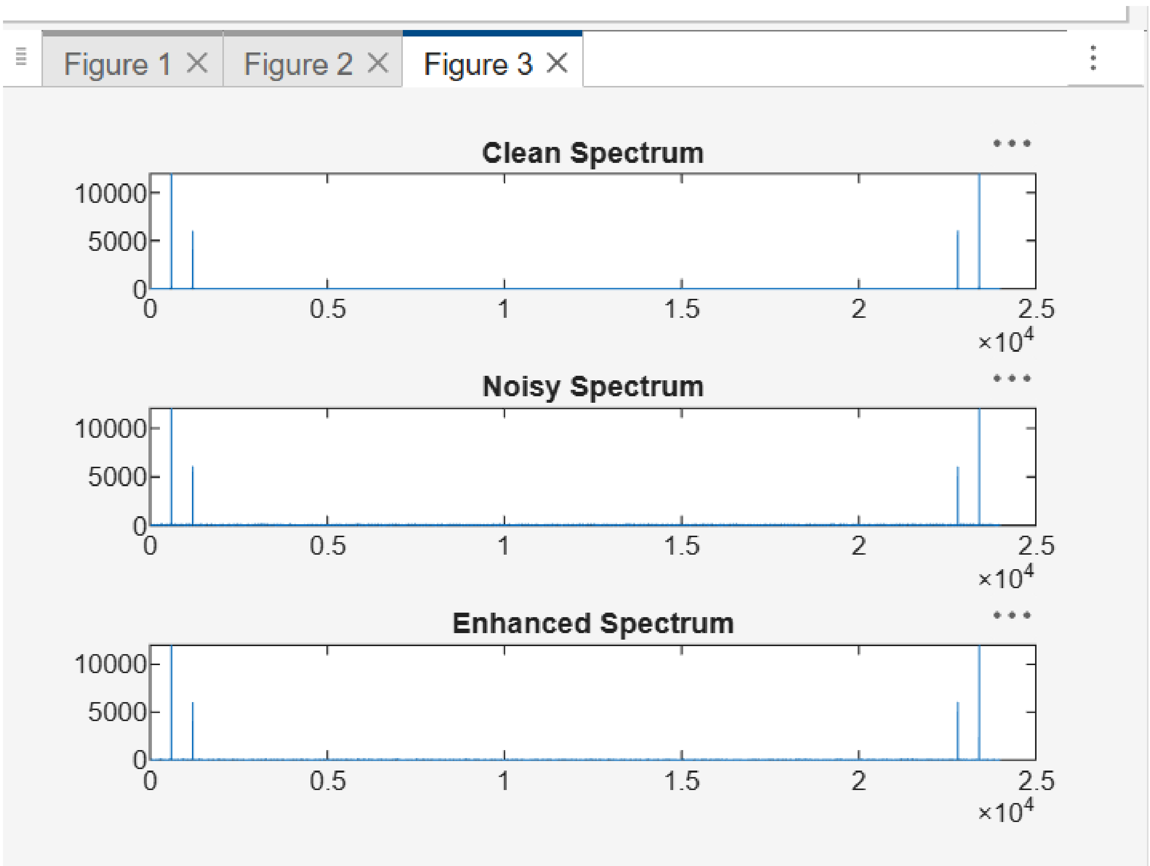


Figure 1
Figure 2
Figure 3
Clean Spectrum
Noisy Spectrum
Enhanced Spectrum
10000
5000
0
0.5
1
1.5
2
2.5
×10^4